SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA0606	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	JOTHIKA M	**Reg. No.:**	192324235
Section / Batch:	SLOT C	**Date:**	28-07-2026

Code of Conduct

I JOTHIKA M (192324235 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Problem 7 (10 Marks)

Scenario: A fitness-tracking app records daily step counts and sleep hours for a user over 30 days, mapped as a scatter plot with Steps on the x-axis, Sleep Hours on the y-axis, and each point sized according to Calories Burned.

a) How many variables are being visually encoded in this chart, and through which aesthetics?

b) Is size an appropriate aesthetic for representing Calories Burned here? What is a limitation of using size?

c) If larger points (higher calories burned) tend to appear at higher step counts but lower sleep hours, what relationship can you infer?

Problem 17 (10 Marks)

Scenario: A bar chart of 12 months' revenue is colored entirely in gray except for the single highest-revenue month, which is colored bright red.

a) What purpose does color serve in this chart — distinguishing, representing value, or highlighting? Justify.

b) Would this technique still be effective if three months were colored red instead of one? Explain why or why not.

c) What is the risk of using this same red highlight color elsewhere in a compans charts inconsistently?

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Interpretation Exercise

CO1 · BT3: Apply ·

This rubric is used to assess student responses to the AT2 Data Interpretation Exercise problems, in which students interpret a described data visualization (data types, aesthetic mappings, scales, coordinate systems, and color scales) and answer accompanying sub-questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough, accurate understanding of data types (nominal, ordinal, numerical), aesthetic mappings, scales, coordinate systems, and color scales as they apply to the given visualization scenario.	Good understanding of the relevant visualization concepts, with only minor gaps in identifying data types, aesthetic mappings, or scale choices.	Basic understanding of visualization concepts, but with some misconceptions about data types, aesthetics, coordinate systems, or color scales.	Poor understanding of core visualization concepts; major misconceptions about data types, aesthetics, coordinate systems, or color scales.
Explanation	**25%**	Provides detailed, well-reasoned explanations for each sub-question, using specific details from the scenario and correct visualization terminology.	Provides clear explanations with adequate justification, referencing the scenario appropriately for most sub-questions.	Explanations are limited or superficial; lacks depth or fails to consistently connect reasoning to the given scenario.	Explanations are poor, unclear, or missing; answers are unsupported assertions with no reasoning shown.
Application	**20%**	Effectively applies visualization concepts to interpret the specific scenario, drawing accurate, well-reasoned conclusions from the described chart or data pattern.	Applies concepts to interpret the scenario correctly, with only minor errors in reasoning or in the conclusions drawn.	Limited application of concepts to the scenario; conclusions are weakly supported or only partially correct.	Fails to apply concepts to the scenario, or the conclusions drawn are incorrect or irrelevant to the data shown.
Organization	**15%**	Answers are well-structured, addressing each sub-question (a, b, c) in a clear, logical sequence with coherent flow throughout.	Answers are organized and address each sub-question, with only minor issues in flow or clarity.	Basic structure; sub-questions are addressed but answers lack clarity or a logical sequence.	Poorly organized; answers are incoherent, and sub-questions are left unaddressed or answered out of order.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Accuracy	**10%**	All parts of the answer — data type identification, aesthetic/scale justification, and final interpretation — are completely accurate.	Mostly accurate, with only minor omissions or a small error in one part of the answer.	Partially correct; some key points (e.g., data type, aesthetic justification, or interpretation) are missing or incorrect.	Answers are largely incorrect, with major gaps in identifying data types, aesthetics, scales, or drawing conclusions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

CO 1 AT 2

DATA INTERPRETATION EXERCISE

NAME: JOTHIKA M

REG.NO: 192324235

COURSE CODE: DSA0606

COURSE NAME: DATA HANDLING AND VISUALIZATION

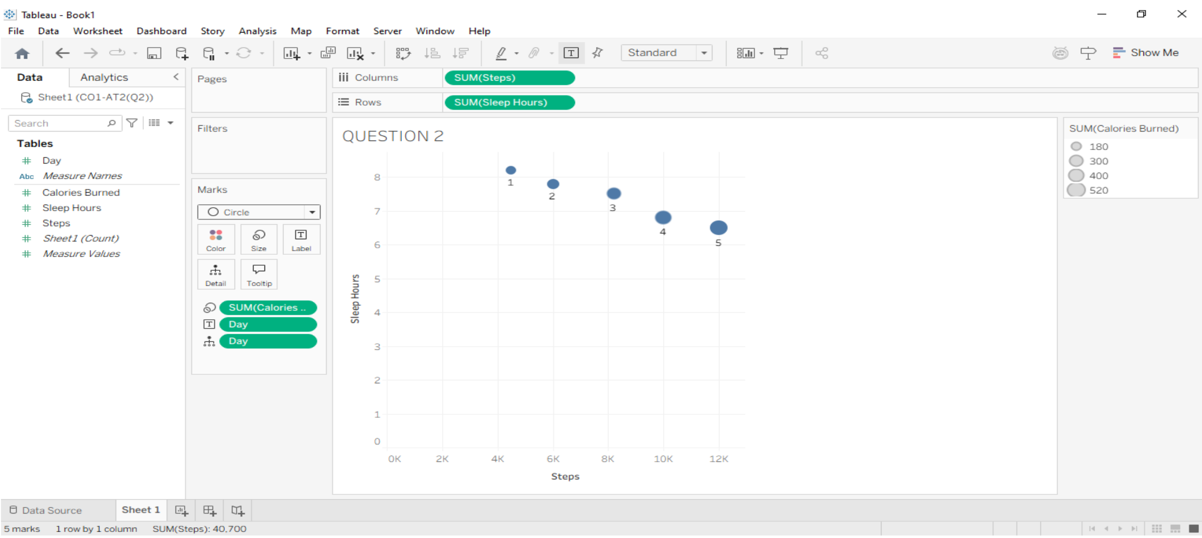
DATE: 28.07.2026

PROBLEM-7

1.Scenario: A fitness-tracking app records daily step counts and sleep hours for a user over 30 days,mapped as a scatter plot with Steps on the x-axis, Sleep Hours on the y-axis, and each point sizedaccording to Calories Burned.

Introduction

A scatter plot is used to study the relationship between two numerical variables by plotting data points on the x-axis and y-axis. In this scenario, the fitness-tracking app records daily **step counts**, **sleep hours**, and **calories burned** over 30 days. The x-axis represents **Steps**, the y-axis represents **Sleep Hours**, and the **size of each point** represents **Calories Burned**. This visualization helps identify trends, patterns, and possible relationships among these variables.



Observation

- **X-axis:** Daily Steps
- **Y-axis:** Sleep Hours
- **Point Size:** Calories Burned
- **Each Point:** One day's fitness data

The chart simultaneously displays three variables:

1. Steps using horizontal position.
2. Sleep hours using vertical position.
3. Calories burned using the size of each point.

Larger circles indicate more calories burned, while smaller circles indicate fewer calories burned.

Evidence

From the scatter plot, it is observed that:

- Larger circles are mostly located at **higher step counts**.
- These larger circles also appear at **lower sleep hours**.
- Smaller circles are generally found at lower step counts with relatively higher sleep hours.

This pattern suggests that days with greater physical activity resulted in higher calorie expenditure.

Interpretation

The visualization indicates a **positive relationship** between **steps and calories burned**, meaning that as the number of steps increases, the calories burned also increase.

It also suggests a **negative relationship** between **steps and sleep hours**, as higher step counts tend to coincide with lower sleep durations.

However, this is only an observed trend from the data and **does not prove a cause-and-effect relationship**. Other factors such as exercise intensity, diet, age, or daily routine may also influence calories burned and sleep hours.

a) How many variables are being visually encoded in this chart, and through which aesthetics?

Introduction:

The scatter plot represents multiple variables using different visual aesthetics.

Content:

- **Steps** → X-axis (Position)
- **Sleep Hours** → Y-axis (Position)
- **Calories Burned** → Size of the points

Interpretation:

The chart encodes **3 variables** using **position (x and y)** and **size**, allowing multiple data values to be viewed in one chart.

b) Is size an appropriate aesthetic for representing Calories Burned here? What is a limitation of using size?

Introduction:

Size is commonly used to represent quantitative values in scatter plots.

Content:

- Yes, size is appropriate because larger points indicate more calories burned.
- Limitation: It is difficult to accurately compare circle sizes, and large points may overlap smaller ones.

Interpretation:

Size helps identify higher calorie values quickly but is less precise than position.

c) If larger points (higher calories burned) tend to appear at higher step counts but lower sleep hours, what relationship can you infer?

Introduction:

The scatter plot helps identify relationships between variables.

Content:

- More **Steps** → More **Calories Burned** (positive relationship).
- More **Steps** → Less **Sleep Hours** (negative relationship).

Interpretation:

Higher physical activity is associated with increased calorie burning and lower sleep duration.

Conclusion

The scatter plot effectively represents three variables in a single visualization by using **position** and **size** as visual aesthetics. It helps users quickly identify relationships between daily steps, sleep hours, and calories burned. The chart suggests that increased physical activity is associated with higher calorie expenditure and relatively lower sleep duration, making it a useful tool for analyzing personal fitness trends and supporting health-related decisions.

PROBLEM-17

2. Scenario: A bar chart of 12 months& revenue is colored entirely in gray except for the single highest- revenue month, which is colored bright red.

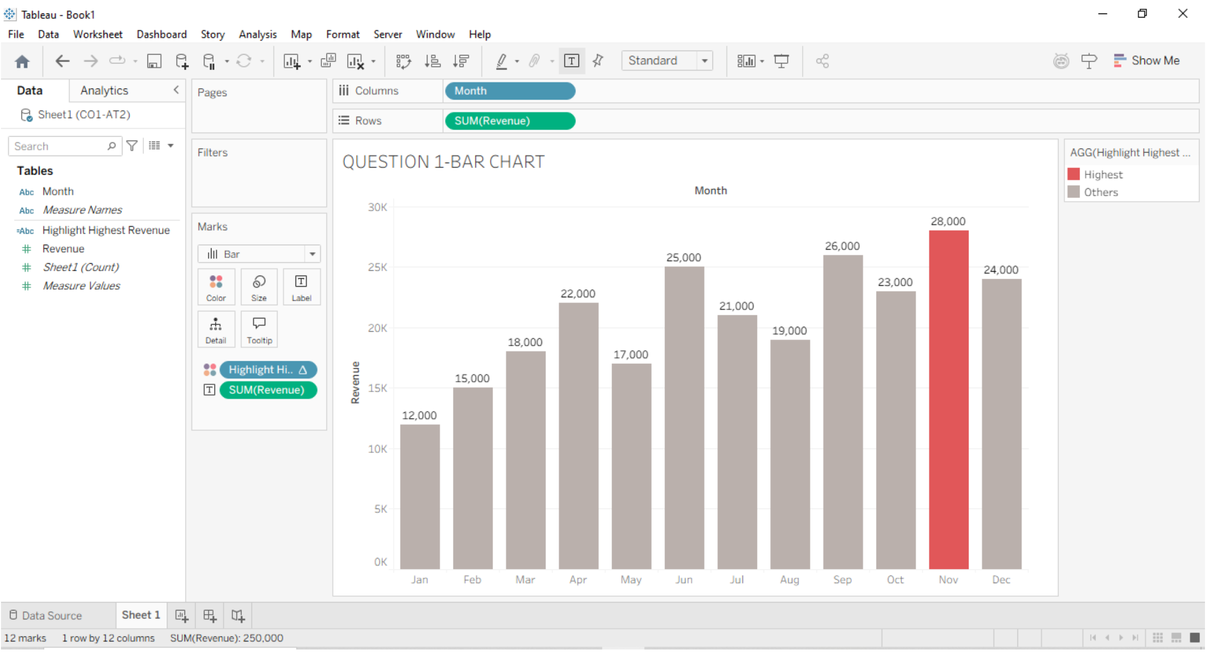
a) What purpose does color serve in this chart — distinguishing, representing value, or highlighting? Justify.

b) Would this technique still be effective if three months were colored red instead of one? Explain why or why not.

c) What is the risk of using this same red highlight color elsewhere in a company& charts inconsistently?

Introduction

This bar chart uses **color** as a visual aesthetic to emphasize important information. All 12 months' revenue is displayed in gray, while the highest-revenue month is highlighted in bright red. This helps viewers quickly identify the most significant data point.



Evidence

The chart contains one red bar representing the highest revenue month, whereas the remaining eleven bars are gray. The use of a contrasting color makes the highlighted month immediately noticeable.

Interpretation

The red color is used for **highlighting**, not for representing revenue values or distinguishing categories. Highlighting only one bar effectively attracts attention to the highest-performing month. However, if multiple bars were colored red or if red were used inconsistently across different charts, it could confuse viewers and reduce the effectiveness of the visualization.

a) What purpose does color serve in this chart — distinguishing, representing value, or highlighting? Justify.

Introduction

Color is a visual aesthetic used to emphasize or communicate information in a chart.

Evidence

The bar chart shows all 12 months in **gray**, while only the **highest-revenue month** is colored**bright red**.

Interpretation

The red color is used for **highlighting** because it immediately draws attention to the highest-revenue month. It is not used to represent different revenue values or different categories.

b) Would this technique still be effective if three months were colored red instead of one? Explain why or why not.

Introduction

Highlighting is most effective when used on a limited number of important data points.

Evidence

If **three months** are colored red instead of one, multiple bars receive the same emphasis.

Interpretation

The viewer's attention is divided among the three months, making it difficult to identify the single highest-revenue month. The visual impact of highlighting is reduced.

c) What is the risk of using this same red highlight color elsewhere in a company's charts inconsistently?

Introduction

Consistent color usage is important for clear and accurate data interpretation.

Evidence

Suppose red indicates **highest revenue** in one chart but represents **loss** or **warning** in another chart.

Interpretation

Different meanings for the same color can confuse viewers and lead to incorrect interpretation of the data.

Conclusion

The chart demonstrates that color is a powerful tool for emphasizing key information when used consistently and sparingly. Proper use of highlighting improves readability and helps users interpret the data accurately.